

# WIRELESS EQUIPMENT CONTROL

## Using MATLAB-Based GUI

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In process instrumentation and control, operators in the plant control room monitor on-field sensors and send real-time commands to control actuators and final control elements such as motors,



Fig. 1: Authors' prototype of wireless equipment control using MATLAB-based GUI

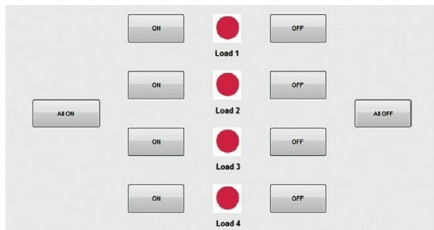


Fig. 2: MATLAB based GUI for controlling four electrical equipment



Fig. 3: Block diagram of the transmitter unit

pumps, variable-speed drives, variable-frequency drives and electrically operated valves in the field using a human-machine interface (HMI) program for process control. A graphical user interface (GUI)-based HMI application enhances the user-friendliness of such an approach.

Signals from the operator station are usually sent to on-field actuators by using an industry-standard 4-20mA current loop protocol. However, due to inherent disadvantages during troubleshooting, control loop testing and plant commissioning stages of a wired communication based process plant, the trend in recent years has been towards operating plants wirelessly. For example, WirelessHART (Highway Addressable Remote Transducer)

protocol based transmitters and actuators are widely used in modern process industries.

In this project, a GUI built on MATLAB environment is employed to wirelessly control four electrical equipment. Transmitter and receiver modules operate at 434MHz frequency. Authors' prototype and the front panel of the MATLAB-based GUI are shown in Figs 1 and 2, respectively.

### Circuit and working

Block diagrams of transmitter and receiver circuits are shown in Figs 3 and 4, respectively. The main components used in this project are described below.

**Arduino Uno.** Arduino Uno is an AVR ATmega328P microcontroller-based development board with six analogue input pins and 14 digital input/output (I/O) pins. The microcontroller has 32kB of ISP flash memory, 2kB RAM and 1kB EEPROM. The board provides serial communication capability via UART, SPI and I2C. The microcontroller can operate at a clock frequency of 16MHz.

In this project, digital I/O pins 8 through 12 of the Arduino are configured as output pins.

**HT12E encoder.** It is a <sup>12</sup> series CMOS LSI encoder for remote control applications. It is capable of encoding information, which consists of 'N' address bits and '12-N'



Fig. 4: Block diagram of the receiver unit